

WinDbg Commands for Exploit Development & Security Research

Privilege Escalation & Token Stealing

- !process 0 0 - List all processes with EPROCESS addresses
- dt _EPROCESS <addr> Token - Dump the Token pointer
- !token - Show the token details of a process
- eq <addr> <value> - Patch memory (e.g., Token pointer)
- .process /r /p <EPROCESS> - Switch debugger context to another process
- !handle - View all handles of a process

Memory Inspection & Exploit Targeting

- dd / dq / dc - Dump memory at a given address
- !pool <addr> - Inspect pool memory
- !vm - Show virtual memory layout
- !address <addr> - Identify memory region of a given address
- !pte <addr> - Page table entry details

Bugcheck / Exception Analysis

- !analyze -v - Analyze bugcheck or crash (full detail)
- .exr -1 / .cxr -1 - View last exception/context records
- !errrec - View WHEA hardware error records

Control Flow / Stack Analysis

- k / kp / kv - Show current call stack
- u <addr> - Disassemble at memory location
- uf <function> - Disassemble full function
- r - View/edit CPU registers
- .ra - Show return address chain

Security Structure Internals

- dt _EPROCESS - View EPROCESS structure
- dt _ETHREAD - View ETHREAD structure
- dt _TOKEN - View security token internals
- !object \Driver - View kernel object namespace
- !drvobj / !devobj - Driver/device inspection

Other Helpful Tools

- .reload /f - Force reload of symbols
- !locks - Show held kernel locks
- !thread - Thread info and wait reason
- !running -ti - Show running threads

Advanced / Dirty Work

- eb / ed / eq - Edit memory directly (byte, dword, qword)
- ba e 1 <addr> - Hardware execution breakpoint
- .writemem <file> <addr> L<size> - Dump memory to file